Fine-Tuning a Small LLM for Customer Support FAQs

Problem Statement:

Generic LLMs like LLaMA, Mistral, or DistilGPT-2 are not domain-specific. This project aims to fine-tune a small LLM (like DistilGPT-2) to generate helpful responses to customer queries based on a small corpus of FAQs from a specific domain (e.g., e-commerce, banking, or SaaS).

The model should learn to generate concise, context-appropriate answers for typical customer questions.

Deliverables:

1. Cleaned and formatted dataset of customer queries and FAQ-style answers.
2. A fine-tuned LLM (checkpoints + inference script).
3. Evaluation results (manual + automated metrics).
4. A short report (README) explaining dataset, training, challenges, and sample outputs.

Sub-Tasks Breakdown:

1. Dataset Collection and Preparation
   - Select a domain (e.g., e-commerce).
   - Collect or synthesize ~500-1000 Q&A pairs.
   - Format into JSONL:

     {"prompt": "How do I track my order?", "response": "You can track your order under 'My Orders'."}

2. Model Selection
   - Choose a small, low-compute model (e.g., DistilGPT-2 or TinyLLaMA).
   - Use Hugging Face Transformers + PEFT or LoRA to minimize training cost.

3. Fine-Tuning
   - Use Hugging Face-s Trainer or trl library.
   - Train on a small scale (e.g., 3-5 epochs, batch size 8).

- Implement early stopping or use validation loss to prevent overfitting.

4. Evaluation

- Manual: Compare base vs fine-tuned model outputs on 10 test queries.

- Automated: Use BLEU / ROUGE / embedding similarity (e.g., cosine on SentenceTransformers).

5. Inference Script

- CLI or Gradio interface to input a customer query and get a model-generated answer.

6. Report

- Describe dataset, training setup, results, and limitations.

Minimal Compute Strategy:

- Use Colab with GPU (T4).

- Model size: <=125M parameters.

- Use 8-bit or 4-bit quantization where applicable.

- LoRA + accelerate for memory-efficient training.

Example Output After Fine-Tuning:

Input: "Can I return a product after 30 days?"

Output: "Returns are only accepted within 30 days of delivery. Sorry!"